Jiayi Chen

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Education

Nanyang Technological University (NTU), Singapore

Doctor of Philosophy, Mechanical and Aerospace Engineering, Expected Graduation: 2030

University of British Columbia (UBC), Canada

Bachelor of Applied Science, Electrical Engineering, 2022-2026

- **Relevant Coursework:** Electrical Circuit Analysis, FPGA, Computer Systems, Data Structure, Linear Algebra, Statistics, Engineering Electromagnetics, Electromechanics, Signal and Systems, System and Control, Machine Learning, Deep Learning, Analog CMOS IC Design, Engineering Design.

Technical Skills

- **Programming Languages:** C, Python, MATLAB
- **Embedded Platforms:** Arduino, STM32, ESP32
- **Circuit Simulation**: LTSPICE, Multisim
- **Reinforcement Learning**: IsaacGym, IsaacSim, Gymnasium, Pybullet; familiar with open-source libraries like RSL_RL (ETH Zurich) and TITA_RL (Direct-Drive Technology)
- **Operating System**: Windows, Linux

Internship

Robotic Motion Control Intern **05/2025-08/2025**

Direct-Drive Technology, Dongguan, China

- Used NVIDIA IsaacGym and IsaacSim to tune reward functions that helped a bipedal wheel-legged robot climb ramps, jump, and recover from falls, while also reorganizing open-source environment documentation to reflect hardware and software updates.

Robotic Motion Control Intern (Remote) **10/2025-12/2025**

Direct-Drive Technology

- Developed motion control algorithms enabling a quadruped robot to climb slopes, navigate stairs, and recover from falls.

Research

Robotic Arm Motion Control with Tactile Sensors, UBC Molecular Mechatronics Lab 09/2025 – 5/2026

- Wrapped a robotic arm with soft tactile sensors to detect contact location and magnitude, used joint states with sensor feedback to train a reinforcement learning policy in Pybullet to learn to detect the unknown environment, then transferred the policy to a physical robot for testing.

Teaching

Teaching Assistant ELEC391, Engineering Design StudioII by Dr. Leo Stocco **01/2026 – 05/2026**

- Assisted students with embedded system design and sensor integration, held weekly lab sessions to troubleshoot circuit and control system implementations, and resolved online feedback on PID controllers and real-time data acquisition.

Academic Projects

Reflow Soldering Control System, UBC **02/2024 - 03/2024**

- Used a PD controller with thermocouple feedback to manage temperature profiles for soldering stages, monitored real-time temperature using Python on a personal computer, and built a breadboard-based controller with an LCD screen for remote control.

Remote Controller Metal Detection Robot, UBC **03/2024 - 05/2024**

- Developed the robot using STM32 and a PD controller in C and built a wireless controller on a breadboard using the JDY40 module.

DJI RoboMaster Robot Development **09/2023 – 5/2025**

- Used UART, I2C, and CAN for robot communication and control, tuned PID controllers for the robot chassis and gimbal, and operated the robot in test fields and competition.

Bipedal Wheel-leg Self-balancing Robot, UBC **01/2025 – 04/2025**

- Implemented a multi-loop PID controller on an ESP32, integrated YOLO-based object detection on an ESP32-Cam, and wrote technical documents, user manuals, and usage instructions.

Machine Learning Model Training and Comparison, UBC **01/2025 – 04/2025**

- Trained and compared common machine learning models (e.g., linear regression, SVM, random forest) using scikit-learn and TensorFlow.

Honors and Awards

- **Outstanding International Student Award - 20,000 CAD, UBC** **2022-2023**
- **International Major Entrance Scholarship - 80,000 CAD, UBC** **2022-2026**
- **UBC ECE Dean's List**

Additional Skills

- Professional proficiency in English.
- Basic video editing skills in Adobe Premiere Pro and After Effects.